Chapter 5
Orthogonality

5.5 Orthonormal Sets

Note on Inner Product Spaces

- In the textbook, when you see the inner product $\langle \mathbf{x}, \mathbf{y} \rangle$ in $\mathbb{R}^n$, it means the scalar product $\mathbf{x}^T\mathbf{y}$.

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T\mathbf{y} \quad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^n)$$

  - For ANY vector space V, there is an abstract definition of inner product $\langle \mathbf{x}, \mathbf{y} \rangle$: it must satisfy

1. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$
2. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$
3. $\langle \alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z} \rangle = \alpha\langle \mathbf{x}, \mathbf{z} \rangle + \beta\langle \mathbf{y}, \mathbf{z} \rangle$

  - $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T\mathbf{y}$ satisfies all 3 properties above.
  - One can define $\langle \cdot, \cdot \rangle$ on other vector spaces such as matrices $\mathbb{R}^{m \times n}$, continuous functions $C[a, b]$, and polynomials P_n. For example, we can also calculate projections and least square solutions in these vector spaces with inner product.

Orthogonal and Orthonormal Set

Definition: A set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ of nonzero vectors is said to be an **orthogonal set** if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ whenever $i \neq j$.

- Any 2 vectors in an orthogonal set are perpendicular.

Definition: An **orthonormal set** of vectors is an orthogonal set of unit vectors.

- $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is orthonormal if and only if

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

- You can turn an orthogonal set into an orthonormal set by scaling each vector to have unit length: $\mathbf{u}_i = \dfrac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$

Example: Orthogonal, Orthonormal

- The set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \ldots, \mathbf{e}_n\}$ in $\mathbb{R}^n$ is orthonormal.
- Is $\left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\2\\-1\end{bmatrix}\right\}$ in $\mathbb{R}^3$ an orthonormal set?
  - $\mathbf{v}_1^T\mathbf{v}_2 = 0, \mathbf{v}_1^T\mathbf{v}_3 = 0, \mathbf{v}_2^T\mathbf{v}_3 = 0$, so this is an orthogonal set.
  - The lengths of the 3 vectors are not 1, so it is not an orthonormal set.
  - To turn it into an orthonormal set, divide each vector by its length:
    $\left\{\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\end{bmatrix}, \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{3}}\begin{bmatrix}-1\\1\\1\end{bmatrix}, \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{1}{\sqrt{6}}\begin{bmatrix}1\\2\\-1\end{bmatrix}\right\}$ is an orthonormal set.

Orthogonal implies Independent

Theorem [Nonzero orthogonal set is linearly independent]
If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is an orthogonal set of nonzero vectors, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ is linearly independent.

Proof

- If $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$, need to show $a_1 = a_2 = \cdots = 0$.
- Taking the inner product of $\mathbf{v}_1$ on both sides:

$$\begin{aligned}&\langle \mathbf{v}_1, a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k \rangle \\ &= a_1\langle \mathbf{v}_1, \mathbf{v}_1 \rangle + a_2\langle \mathbf{v}_1, \mathbf{v}_2 \rangle + \cdots + a_k\langle \mathbf{v}_1, \mathbf{v}_k \rangle \quad \text{orthogonal} \\ &= a_1\langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 0\end{aligned}$$

but since $\mathbf{v}_1 \neq \mathbf{0}, \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = \|\mathbf{v}_1\|^2 > 0$, so $a_1 = 0$.

- We can similarly show $a_i = 0$ for any i, by taking the inner product of $\mathbf{v}_i$.
- Therefore, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ is linearly independent.

Orthonormal Basis

Theorem [Nonzero orthogonal set is linearly independent]
If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is an orthogonal set of nonzero vectors, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ is linearly independent.

Consequence

- If $B = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is an orthonormal set, then by the theorem above, B is a basis for the subspace $S = \mathrm{Span}(\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k) = R(U)$.
- We say that B is an **orthonormal basis** for S.
- Orthonormal basis makes many calculations much simpler:
  - Coordinate vector, inner product, length
  - Least squares solution, projection

Example: orthonormal basis

- The set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$ is an orthonormal basis for $\mathbb{R}^4$.
- How about this one?

$$\left\{\begin{bmatrix} 1/2 \\ -1/2 \\ -1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ -1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}\right\}$$

Yes, it is an orthonormal basis for $\mathbb{R}^4$

- $\left\{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{3}}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}\right\}$ is an orthonormal basis for the plane $S = \text{Span}([1,0,1]^T, [-1,1,1]^T)$.

Orthogonal matrix

Definition

An $n \times n$ matrix Q is called an orthogonal matrix if the columns of Q form an orthonormal basis in $\mathbb{R}^n$.

Orthogonal matrices must be square $n \times n$ (basis of

Examples

(From earlier slides, just in matrix form)

- $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & 2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & -1/\sqrt{6} \end{bmatrix}$
- $Q = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

Equation for Orthogonal Matrix

Theorem

An $n \times n$ matrix Q is an orthogonal matrix if and only if

$$Q^T Q = I_n$$

Proof

$n \times n$ matrix Q is orthogonal

$\Leftrightarrow$ Column vectors of Q satisfy

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \mathbf{q}_i^T \mathbf{q}_j = \delta_{ij}$$

(However, $\mathbf{q}_i^T \mathbf{q}_j$ is the (i, j) entry of $Q^T Q$, and δ_{ij} is the (i, j) entry of I_n.)

$\Leftrightarrow Q^T Q = I_n$

$\Leftrightarrow Q$ is nonsingular and $Q^{-1} = Q^T$

Orthogonal Matrix Properties

Let Q be an $n \times n$ matrix. The following are equivalent:

- Q is an orthogonal matrix.
- The columns of Q form an orthonormal basis for $\mathbb{R}^n$.
- $Q^T Q = I$, that is, $Q^T = Q^{-1}$.
- $(Q\mathbf{x})^T (Q\mathbf{y}) = \mathbf{x}^T Q^T Q\mathbf{y} = \mathbf{x}^T \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.
- $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{R}^n$.

Orthonormal Sets and Least Squares

Recall: If A is an $m \times n$ matrix of rank n (independent columns, $m > n$), then the least squares solution to $A\mathbf{x} = \mathbf{b}$ is

$$A^T A\hat{\mathbf{x}} = A^T\mathbf{b} \Rightarrow \hat{\mathbf{x}} = \left(A^T A\right)^{-1} A^T\mathbf{b}$$

Having an orthonormal set makes $A^T A$ much simpler!

Theorem

If the column vectors of $m \times n$ matrix A form an orthonormal set of vectors in $\mathbb{R}^m$, then $A^T A = I_n$ and the solution to the least squares problem $A\mathbf{x} = \mathbf{b}$ is

$$\hat{\mathbf{x}} = A^T\mathbf{b}$$

Example: Least Squares

Find the solution to the least squares problem $A\mathbf{x} = \mathbf{b}$,

$$\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{3} \\ 0 & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Solution

The column vectors of A form an orthonormal set.

Therefore, the least squares solution is simply

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4/\sqrt{2} \\ 4/\sqrt{3} \end{bmatrix}$$

Orthonormal Sets and Projection

Recall: If $\hat{\mathbf{x}}$ is a least squares solution to $A\mathbf{x} = \mathbf{b}$, then $\mathbf{p} = A\hat{\mathbf{x}}$ is the unique point on $S = R(A)$ that is closest to $\mathbf{b}$. Such $\mathbf{p}$ is called the **projection** of $\mathbf{b}$ onto $S = R(A)$.

Theorem If the column vectors of $m \times n$ matrix A form an orthonormal set of vectors in $\mathbb{R}^m$, then the projection of $\mathbf{b} \in \mathbb{R}^m$ onto $S = R(A)$ is

$$\mathbf{p} = A\hat{\mathbf{x}} = AA^T\mathbf{b}$$

If $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is an orthonormal basis for a nonzero subspace $S = R(U)$ of $\mathbb{R}^m$, then the projection $\mathbf{p}$ of $\mathbf{b} \in \mathbb{R}^m$ onto S is $\mathbf{p} = UU^T\mathbf{b}$

$P = UU^T$ is the projection matrix corresponding to subspace S.